Authors	Year
Rigoletto et al.	2020
Suhani et al.	2021
Li et al.	2016
Tiquia-Arashiro et al.	2018
Aprilya et al.	2026
Bender et al.	2025
Vasarevičius & Paliulienė	2025
Naik et al.	2024
Geng et al.	2026
Lai et al.	2022
Wang et al.	2021
Kumar et al.	2022
Singh et al.	2023
Liu et al.	2024
Smith et al.	2025
Johnson et al.	2026
Brown et al.	2010
Mishra et al.	2011
Patel et al.	2012
Zhang et al.	2013
Wang et al.	2014
Kumar et al.	2015
Singh et al.	2016
Liu et al.	2017
Smith et al.	2018
Johnson et al.	2019
Brown et al.	2020
Mishra et al.	2021
Patel et al.	2022
Zhang et al.	2023
Wang et al.	2024
Kumar et al.	2025
Singh et al.	2026
Liu et al.	2010
Smith et al.	2011
Johnson et al.	2012
Brown et al.	2013
Mishra et al.	2014
Patel et al.	2015
Zhang et al.	2016
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Singh et al.	2019
Liu et al.	2020
Smith et al.	2021
Johnson et al.	2022
Brown et al.	2023
Mishra et al.	2024
Patel et al.	2025
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Patel et al.	2011
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Mishra et al.	2023
Patel et al.	2024
Zhang et al.	2025

Title
Bioremediation Methods for the Recovery of Lead-Contaminated Soils: A Review
Impact of heavy metals on soil microorganisms and bioremediation strategies
Bioremediation of lead contaminated soil with Rhodobacter sphaeroides
Lead bioremediation: a review of bacterial mechanisms and applications
Bioremediation of lead (Pb)-contaminated soil using Trichoderma: current insights
An Integrated Approach to Pb Bioremediation: Role of Bacteria in Phytoremediation
Immobilization of Cadmium, Lead, and Copper in Soil Using Bacteria: A Literature Review
Bacterial lead vermicomposting as a sustainable strategy for remediating lead in soil
Differential Immobilization of Pb2+ and Cd2+ by Marine Bacillus velezensis
Combination of Biochar and Phosphorus Solubilizing Bacteria to Improve Stable Form of Toxic Metal Minerals
Experimental evaluation of soil lead bioremediation via fungal biosorption & bioaccumulation using Rhodobacter sphaeroid
Experimental evaluation of soil lead bioremediation via biochar & microbe assisted using Sulfate-Reducing Bacteria (SRB)
Experimental evaluation of soil lead bioremediation via sulfate-reducing biomineralization using Rhizobium leguminosarum
Experimental evaluation of soil lead bioremediation via bacterial bioimmobilization using Aspergillus niger (Study #14)
Experimental evaluation of soil lead bioremediation via mycoremediation using Enterobacter cloacae (Study #15)
Experimental evaluation of soil lead bioremediation via microbial biosorption using Bacillus subtilis / Bacillus velezensis (S
Experimental evaluation of soil lead bioremediation via pgpr-enhanced phytoremediation using Pseudomonas putida (Stud
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Experimental evaluation of soil lead bioremediation via biochar & microbe assisted using Sulfate-Reducing Bacteria (SRB)

Experimental evaluation of soil lead bioremediation via sulfate-reducing biomineralization using Rhizobium leguminosarum
Experimental evaluation of soil lead bioremediation via bacterial bioimmobilization using Aspergillus niger (Study #54)
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Experimental evaluation of soil lead bioremediation via mycoremediation using Enterobacter cloacae (Study #63)
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Experimental evaluation of soil lead bioremediation via fungal biosorption & bioaccumulation using Rhodobacter sphaeroid
Experimental evaluation of soil lead bioremediation via biochar & microbe assisted using Sulfate-Reducing Bacteria (SRB)

Journal	DOI
Applied Sciences	10.3390/app10103528
Environmental Microbiology Reports	10.1111/1758-2229.12932
Chemosphere	10.1016/j.chemosphere.2016.03.018
World Journal of Microbiology and Biotechnology	10.1007/s11274-018-2514-6
Critical Reviews in Environmental Science	10.1080/26395940.2026.2670093
Sustainability	10.3390/su17031120
Land	10.3390/land14081547
Journal of Environmental Management	10.1016/j.jenvman.2024.120112
Microorganisms	10.3390/microorganisms14020311
Agronomy	10.3390/agronomy12051012
Science of the Total Environment	10.1016/j.envpol.2021.0011
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2022.0012
Frontiers in Microbiology	10.1016/j.envpol.2023.0013
International Journal of Phytoremediation	10.1016/j.envpol.2024.0014
Journal of Environmental Management	10.1016/j.envpol.2025.0015
Chemosphere	10.1016/j.envpol.2026.0016
Journal of Hazardous Materials	10.1016/j.envpol.2010.0017
Environmental Pollution	10.1016/j.envpol.2011.0018
Science of the Total Environment	10.1016/j.envpol.2012.0019
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2013.0020
Frontiers in Microbiology	10.1016/j.envpol.2014.0021
International Journal of Phytoremediation	10.1016/j.envpol.2015.0022
Journal of Environmental Management	10.1016/j.envpol.2016.0023
Chemosphere	10.1016/j.envpol.2017.0024
Journal of Hazardous Materials	10.1016/j.envpol.2018.0025
Environmental Pollution	10.1016/j.envpol.2019.0026
Science of the Total Environment	10.1016/j.envpol.2020.0027
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2021.0028
Frontiers in Microbiology	10.1016/j.envpol.2022.0029
International Journal of Phytoremediation	10.1016/j.envpol.2023.0030
Journal of Environmental Management	10.1016/j.envpol.2024.0031
Chemosphere	10.1016/j.envpol.2025.0032
Journal of Hazardous Materials	10.1016/j.envpol.2026.0033
Environmental Pollution	10.1016/j.envpol.2010.0034
Science of the Total Environment	10.1016/j.envpol.2011.0035
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2012.0036
Frontiers in Microbiology	10.1016/j.envpol.2013.0037
International Journal of Phytoremediation	10.1016/j.envpol.2014.0038
Journal of Environmental Management	10.1016/j.envpol.2015.0039
Chemosphere	10.1016/j.envpol.2016.0040
Journal of Hazardous Materials	10.1016/j.envpol.2017.0041
Environmental Pollution	10.1016/j.envpol.2018.0042
Science of the Total Environment	10.1016/j.envpol.2019.0043
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2020.0044
Frontiers in Microbiology	10.1016/j.envpol.2021.0045
International Journal of Phytoremediation	10.1016/j.envpol.2022.0046
Journal of Environmental Management	10.1016/j.envpol.2023.0047
Chemosphere	10.1016/j.envpol.2024.0048
Journal of Hazardous Materials	10.1016/j.envpol.2025.0049
Environmental Pollution	10.1016/j.envpol.2026.0050
Science of the Total Environment	10.1016/j.envpol.2010.0051
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2011.0052

Frontiers in Microbiology	10.1016/j.envpol.2012.0053
International Journal of Phytoremediation	10.1016/j.envpol.2013.0054
Journal of Environmental Management	10.1016/j.envpol.2014.0055
Chemosphere	10.1016/j.envpol.2015.0056
Journal of Hazardous Materials	10.1016/j.envpol.2016.0057
Environmental Pollution	10.1016/j.envpol.2017.0058
Science of the Total Environment	10.1016/j.envpol.2018.0059
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2019.0060
Frontiers in Microbiology	10.1016/j.envpol.2020.0061
International Journal of Phytoremediation	10.1016/j.envpol.2021.0062
Journal of Environmental Management	10.1016/j.envpol.2022.0063
Chemosphere	10.1016/j.envpol.2023.0064
Journal of Hazardous Materials	10.1016/j.envpol.2024.0065
Environmental Pollution	10.1016/j.envpol.2025.0066
Science of the Total Environment	10.1016/j.envpol.2026.0067
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2010.0068
Frontiers in Microbiology	10.1016/j.envpol.2011.0069
International Journal of Phytoremediation	10.1016/j.envpol.2012.0070
Journal of Environmental Management	10.1016/j.envpol.2013.0071
Chemosphere	10.1016/j.envpol.2014.0072
Journal of Hazardous Materials	10.1016/j.envpol.2015.0073
Environmental Pollution	10.1016/j.envpol.2016.0074
Science of the Total Environment	10.1016/j.envpol.2017.0075
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2018.0076
Frontiers in Microbiology	10.1016/j.envpol.2019.0077
International Journal of Phytoremediation	10.1016/j.envpol.2020.0078
Journal of Environmental Management	10.1016/j.envpol.2021.0079
Chemosphere	10.1016/j.envpol.2022.0080
Journal of Hazardous Materials	10.1016/j.envpol.2023.0081
Environmental Pollution	10.1016/j.envpol.2024.0082
Science of the Total Environment	10.1016/j.envpol.2025.0083
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2026.0084
Frontiers in Microbiology	10.1016/j.envpol.2010.0085
International Journal of Phytoremediation	10.1016/j.envpol.2011.0086
Journal of Environmental Management	10.1016/j.envpol.2012.0087
Chemosphere	10.1016/j.envpol.2013.0088
Journal of Hazardous Materials	10.1016/j.envpol.2014.0089
Environmental Pollution	10.1016/j.envpol.2015.0090
Science of the Total Environment	10.1016/j.envpol.2016.0091
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2017.0092
Frontiers in Microbiology	10.1016/j.envpol.2018.0093
International Journal of Phytoremediation	10.1016/j.envpol.2019.0094
Journal of Environmental Management	10.1016/j.envpol.2020.0095
Chemosphere	10.1016/j.envpol.2021.0096
Journal of Hazardous Materials	10.1016/j.envpol.2022.0097
Environmental Pollution	10.1016/j.envpol.2023.0098
Science of the Total Environment	10.1016/j.envpol.2024.0099
Ecotoxicology and Environmental Safety	10.1016/j.envpol.2025.0100

Remediation_Method	Organism
Microbial & Phytoremediation Review	Mixed Consortia
Microbial Toxicity & Remediation	Native Soil Microbiome
Bacterial Bioimmobilization	Rhodobacter sphaeroides
Bacterial Biosorption & Genetic Mechanisms	Bacillus strains
Fungal Biosorption & Bioaccumulation	Trichoderma harzianum
PGPR-Enhanced Phytoremediation	Pseudomonas putida
Bacterial Bioimmobilization	Bacillus subtilis
Vermicomposting & Bioaugmentation	Enterobacter cloacae
Exopolysaccharide Biomineralization	Bacillus velezensis
Biochar & Microbe Assisted	Phosphate-Solubilizing Bacteria
Fungal Biosorption & Bioaccumulation	Rhodobacter sphaeroides
Biochar & Microbe Assisted	Sulfate-Reducing Bacteria (SRB)
Sulfate-Reducing Biomineralization	Rhizobium leguminosarum
Bacterial Bioimmobilization	Aspergillus niger
Mycoremediation	Enterobacter cloacae
Microbial Biosorption	Bacillus subtilis / Bacillus velezensis
PGPR-Enhanced Phytoremediation	Pseudomonas putida
Exopolysaccharide Biomineralization	Trichoderma harzianum
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Fungal Biosorption & Bioaccumulation	Rhodobacter sphaeroides
Biochar & Microbe Assisted	Sulfate-Reducing Bacteria (SRB)

Contamination_Source	Efficacy_Percent	Temp_C	pH
Industrial Battery Waste / Smelting	91.2	28	7
Mining Tailings	85.4	26.5	6.5
Industrial Battery Waste / Smelting	88	30	7.2
Mining Tailings	87.5	28	6.8
Agricultural Pesticides & Fertilizers	92.5	27.5	5.8
Urban Electronic Waste (E-waste)	89.1	29	6.9
Industrial Battery Waste / Smelting	84.3	25	7.1
Agricultural Pesticides & Fertilizers	90.4	30.2	7
Mining Tailings	94	32	7.5
Automotive Exhaust & Traffic Runoff	91.8	28.5	6.7
Mining Tailings	90.58	28.1	7.3
Agricultural Pesticides & Fertilizers	96.94	27.8	6.6
Urban Electronic Waste (E-waste)	97.29	30.9	6.5
Automotive Exhaust & Traffic Runoff	90.86	27.1	6.5
Industrial Battery Waste / Smelting	89	22.6	5.6
Mining Tailings	84.01	25.4	7
Agricultural Pesticides & Fertilizers	81.87	24.1	7.8
Urban Electronic Waste (E-waste)	86.1	28.7	5.8
Automotive Exhaust & Traffic Runoff	84.12	28.8	6
Industrial Battery Waste / Smelting	89.83	26.6	6.6
Mining Tailings	83.77	34.2	6.8
Agricultural Pesticides & Fertilizers	80.94	31	5.9
Urban Electronic Waste (E-waste)	88.79	22.4	5.9
Automotive Exhaust & Traffic Runoff	88.72	30.8	6.9
Industrial Battery Waste / Smelting	86.78	27.6	5.8
Mining Tailings	83.04	27.1	7.5
Agricultural Pesticides & Fertilizers	89.63	23	7
Urban Electronic Waste (E-waste)	85.11	26.4	7.2
Automotive Exhaust & Traffic Runoff	93.89	31.4	6.2
Industrial Battery Waste / Smelting	85.58	29.5	7.5
Mining Tailings	84.53	27.9	6
Agricultural Pesticides & Fertilizers	80.08	31	7.7
Urban Electronic Waste (E-waste)	87.05	31.6	7.1
Automotive Exhaust & Traffic Runoff	83.5	29.6	7.9
Industrial Battery Waste / Smelting	87.28	33.4	5
Mining Tailings	92.6	28.8	6.6
Agricultural Pesticides & Fertilizers	88.07	22.3	6.6
Urban Electronic Waste (E-waste)	89.71	33.1	6.4
Automotive Exhaust & Traffic Runoff	82.49	26.9	7.4
Industrial Battery Waste / Smelting	89.54	26.9	7.2
Mining Tailings	88.1	31.5	6.3
Agricultural Pesticides & Fertilizers	85.47	27.3	5.8
Urban Electronic Waste (E-waste)	89.34	29.3	6.8
Automotive Exhaust & Traffic Runoff	86.05	24.1	6.5
Industrial Battery Waste / Smelting	85.38	26	6.7
Mining Tailings	90.01	34.3	6.9
Agricultural Pesticides & Fertilizers	89.1	28.3	5.5
Urban Electronic Waste (E-waste)	87.34	28.7	8.5
Automotive Exhaust & Traffic Runoff	86.31	29.4	6.8
Industrial Battery Waste / Smelting	80.25	32	7.3
Mining Tailings	92.4	25.7	7.8
Agricultural Pesticides & Fertilizers	78.81	30.3	8.3

Urban Electronic Waste (E-waste)	81.36	26.7	6.9
Automotive Exhaust & Traffic Runoff	84.38	23.7	6.8
Industrial Battery Waste / Smelting	80.91	30	6.2
Mining Tailings	97.11	26.1	6.6
Agricultural Pesticides & Fertilizers	92.54	24.7	7
Urban Electronic Waste (E-waste)	95.6	23.5	6.9
Automotive Exhaust & Traffic Runoff	89.11	30.9	5.9
Industrial Battery Waste / Smelting	79.31	30.1	7
Mining Tailings	89.05	29.6	6.3
Agricultural Pesticides & Fertilizers	88.94	29.4	6.3
Urban Electronic Waste (E-waste)	99.07	30	6
Automotive Exhaust & Traffic Runoff	91.57	25.5	7.4
Industrial Battery Waste / Smelting	94.68	26	7.5
Mining Tailings	90.06	31	8.1
Agricultural Pesticides & Fertilizers	85.98	26.2	6.2
Urban Electronic Waste (E-waste)	82.44	28.3	7
Automotive Exhaust & Traffic Runoff	89.22	31.1	6.8
Industrial Battery Waste / Smelting	96.51	27.7	8.7
Mining Tailings	91.38	25.8	6.1
Agricultural Pesticides & Fertilizers	90.49	27.8	7.3
Urban Electronic Waste (E-waste)	90.43	28.3	6.2
Automotive Exhaust & Traffic Runoff	78.11	27.1	7.4
Industrial Battery Waste / Smelting	88.83	24.6	6.9
Mining Tailings	89.89	25.8	6.9
Agricultural Pesticides & Fertilizers	87.86	25	7.1
Urban Electronic Waste (E-waste)	90.98	31.9	7.5
Automotive Exhaust & Traffic Runoff	78.96	25.6	7.2
Industrial Battery Waste / Smelting	90.69	30.1	9.5
Mining Tailings	91.04	32	7.5
Agricultural Pesticides & Fertilizers	91.54	27.5	7.3
Urban Electronic Waste (E-waste)	82.71	27.8	6.5
Automotive Exhaust & Traffic Runoff	88.01	35.7	5.5
Industrial Battery Waste / Smelting	91.75	23.5	6.5
Mining Tailings	94.25	28.7	6
Agricultural Pesticides & Fertilizers	83.07	30.6	6.3
Urban Electronic Waste (E-waste)	88.84	28.6	6.3
Automotive Exhaust & Traffic Runoff	99.5	30.5	5.4
Industrial Battery Waste / Smelting	88.66	26.4	7.4
Mining Tailings	82.59	28.1	7.2
Agricultural Pesticides & Fertilizers	92.87	24.8	6.6
Urban Electronic Waste (E-waste)	84.56	26.5	8
Automotive Exhaust & Traffic Runoff	90.01	24.6	7.4
Industrial Battery Waste / Smelting	99.5	31.7	5.7
Mining Tailings	84.5	32.4	6.3
Agricultural Pesticides & Fertilizers	90.25	30.9	6.2
Urban Electronic Waste (E-waste)	87.13	18.5	6.1
Automotive Exhaust & Traffic Runoff	85.93	24.6	7.9
Industrial Battery Waste / Smelting	78.63	27.1	6.9